

1. Problem statement & motivation

- For many college students, studying can be difficult. Often, their time has to be split among many classes, and the strategies they use could be deemed ineffective. Students often rely on generic study advice or static schedules that do not account for individual learning styles or course difficulty. As a result, students may waste their time studying in ways that do not fully help them understand concepts, leading to frustration, burnout, and unsatisfactory grades.
- The target users are college students, particularly those new to the demanding workload, those taking various classes, and students looking to improve their efficiency and grades.
- Agentic AI is appropriate in this case because it can perform ongoing planning, monitoring, evaluation, and adaptation. Agentic AI can dynamically plan, record outcomes, and revise strategies, unlike other systems.

2. Use cases & scenarios

- Scenario 1: A student uploads a course syllabus and specifies available study hours. The system generates a personalized study plan and assigns appropriate study methods.
- Scenario 2: The student provides feedback indicating certain study methods are ineffective. The system analyzes logged feedback and adapts future recommendations.

- Scenario 3: The student asks which study methods work best for them. The system summarizes historical data and provides personalized learning insights.

3. Initial system design

Multi-Agent Architecture:

1. Planner agent
 - a. Creates weekly and daily study schedules
 - b. Allocates time for courses and topics
2. Study strategy agent
 - a. Explains why a study method is recommended
 - b. Presents alternative popular study methods such as active recall, spaced repetition, pomodoro, flashcards, and practice problems
3. Feedback and logging agent
 - a. Collects user input after each study session
 - b. Logs metrics such as time spent and effectiveness
4. Performance analyzer agent
 - a. Analyzes logged data over time
 - b. Identifies which study methods are most effective for the user
5. Critic /optimizer agent
 - a. Reviews study plans
 - b. Adjusts workloads, methods, and time allocations to improve effectiveness

Tools, APIs, and Data Sources

- Google opal

- Local storage (json) for user logs and preferences
- File upload for syllabi or course outlines
- Internal evaluation logic for strategy effectiveness

4. Feasibility & risks

- Potential Challenges

- Unrealistic study plans
- Limited evaluation data
- User engagement

- Constraints

- Privacy considerations restricting sensitive data collection
- No access to official grades or learning management systems